

QTM 220 Final Practice Problems

Problem 1

Suppose you have a sample $Y_1, Y_2, \dots, Y_n$ drawn with replacement from a population with mean μ and standard deviation σ .

Consider the following estimator for μ : $\tilde{Y} = \frac{1}{n+2} \sum_{i=1}^n Y_i$

Part A

What is $E[\tilde{Y}]$?

$$\begin{aligned} E[\tilde{Y}] &= E\left[\frac{1}{n+2} \sum_{i=1}^n Y_i\right] \\ &= \frac{1}{n+2} E\left[\sum_{i=1}^n Y_i\right] \\ &= \frac{1}{n+2} \sum_{i=1}^n E[Y_i] \end{aligned}$$

by linearity of expectations

$$= \frac{1}{n+2} \sum_{i=1}^n E[Y]$$

by IID

$$= \frac{1}{n+2} \sum_{i=1}^n \mu$$

by definition

$$= \frac{1}{n+2} n\mu$$

$$= \frac{n}{n+2} \mu$$

Part B

How does $E[\tilde{Y}]$ compare to $E[\bar{Y}]$? (where $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$)

similar steps show that $E[\bar{Y}] = \mu$. $\bar{Y}$ is unbiased, $\tilde{Y}$ is biased, but $\lim_{n \rightarrow \infty} \frac{n}{n+2} = 1$ so bias will decrease as n increases.

Part C

Showing your work, calculate $Var[\tilde{Y}]$.

$$Var[\tilde{Y}] = V[\frac{1}{n+2} \sum_{i=1}^n Y_i]$$

$$= (\frac{1}{n+2})^2 V[\sum_{i=1}^n Y_i]$$

$$= (\frac{1}{n+2})^2 \sum_{i=1}^n V[Y]$$

because IID, covariances are 0

$$= (\frac{1}{n+2})^2 \sum_{i=1}^n \sigma^2$$

by definition

$$= (\frac{1}{n+2})^2 n\sigma^2$$

$$= \frac{n}{(n+2)^2} \sigma^2$$

Part D

How does $\text{Var}[\tilde{Y}]$ compare to $\text{Var}[\bar{Y}]$?

Similar steps show that $V[\tilde{Y}] = \frac{\sigma^2}{n}$

$\frac{1}{n} > \frac{n}{(n+2)^2}$ so our $\tilde{Y}$ estimator has a lower variance than the mean. But the estimator is biased, so we probably wouldn't want to use it over the mean.